AHMED AL-GHOSAINI

M.Sc. Computer Science | Robot Learning · VLA & Foundation Models

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WORK EXPERIENCE

Jan 2026 – Sep 2026
Hannover, Germany

Institute of Mechatronic Systems (imes), Leibniz University Hannover
Master's Thesis Student | VLA Model for Humanoid Robot Navigation

- Fine-tuned a Vision-Language-Action foundation model (OpenVLA) via imitation learning for language-conditioned humanoid navigation in dynamic environments, with the policy predicting action chunks.
- Fused real-time LiDAR and depth streams for obstacle avoidance, processing the sensor data concurrently in a multithreaded pipeline, and integrated the policy into the ROS2 software stack.
- Collected 300 robot episodes via teleoperation and built the multimodal dataset (camera video, robot state, LiDAR/depth, task outcomes).
- Built the complete pipeline end to end (data collection, preprocessing, PyTorch training, real-time inference, deployment on the physical robot), achieving a 90% success rate on real-world tasks.

Sep 2024 – Sep 2026
Berlin, Germany
(Remote)

PwC Germany
Working Student | Data Science & Applied AI Engineering

- Benchmarked LLMs, multimodal models and retrieval methods, and ran ML experiments for image and table processing.
- Co-built an internal Retrieval-Augmented Generation (RAG) auditing tool, cutting manual auditing time by 50–60%.
- Engineered automated, end-to-end document processing pipelines for AI-driven report generation.
- Shipped production Python on Linux/Docker with Git and CI/CD workflow.

Jun 2024 – Sep 2024
Hannover, Germany

L3S Research Center
Research Assistant | Robot Learning & Tactile Sensing

- Ran physical-robot experiments to collect multimodal and tactile/force-sensing datasets in the *VibroSense* project.
- Validated robot-learning algorithms, including reinforcement learning, and worked with simulations (MuJoCo and Isaac Gym).
- Set up and maintained the lab's shared code repositories and their documentation (Git).

Jan 2024 – Dec 2024
Hannover, Germany

Institute for Information Processing (TNT), Leibniz University Hannover
Research Assistant | Deep Learning & Generative Models

- Implemented and trained diffusion-model and CNN architectures in PyTorch (models, training loops, data loaders) for super-resolution of human genome data, achieving results that exceeded state-of-the-art models.
- Designed the preprocessing and evaluation pipelines enabling systematic comparison across model variants within the institute's *genomic data science* project.

Dec 2023 – May 2024
Hannover, Germany

luhbots – RoboCup@Work Team, Leibniz University Hannover
Software Developer | Robotics & Computer Vision

- Co-developed software for an autonomous mobile manipulator competing in *RoboCup@Work* (industrial pick-and-place).
- Improved the robot's autonomous navigation and vision-based object-recognition components on the real platform.

Jun 2023 – Nov 2023
Hannover, Germany

Institute of Production Engineering and Machine Tools (IFW), Leibniz University Hannover
Student Assistant | Machine Learning for Manufacturing

- Built classification and regression models predicting tool wear and process faults in drilling and milling from real machine sensor data, including the raw-sensor preprocessing and augmentation pipeline.

Dec 2021 – Dec 2022
Hannover, Germany

Volkswagen
Working Student | Automobile Production

Assembled vehicle components on the series production line and operated the associated machines and systems (hands-on industrial operations experience).

EDUCATION

Oct 2023 – Sep 2026
Hannover, Germany

Computer Science, Master of Science
Leibniz University Hannover

- **Focus**: Robotics, Computer Vision and Artificial Intelligence.
- **Current Grade**: 1.2 (A+).
- **Thesis**: Vision-Language-Action (VLA) Model for Humanoid Robot Navigation.
- **Relevant Courses**: Machine Learning (1.0), Computer Vision (1.0), Deep Learning (1.0), Robotik I (1.3), Statistical Natural Language Processing (1.0).
- **Systems Coursework** (B.Sc./M.Sc.): Operating Systems, Computer Architecture, Parallel Computing, C++ and Java (OOP).

Apr 2026 – Sep 2026
Tokyo, Japan

Exchange Semester
Tokyo Metropolitan University

Apr 2025 – Sep 2025
Naples, Italy

Exchange Semester
Università degli Studi di Napoli Federico II

Oct 2019 – Sep 2023
Hannover, Germany

Mechatronics, Bachelor of Science
Leibniz University Hannover

- **Final Grade**: 1.9 (in the Top 8% of students).
- **Thesis**: GAN-based deep super-resolution for genomic data, graded 1.0 (A+).

SKILLS

Robot Learning: VLA models, imitation learning, action chunking, reinforcement learning, diffusion models
Robotics: ROS/ROS2, Simulators (MuJoCo, NVIDIA Isaac Gym/Sim)
Programming Languages: Python, C/C++, Java, SQL, Matlab
ML Frameworks: PyTorch, Hugging Face Transformers, TensorFlow, Keras, scikit-learn, Pandas
MLOps: Git, Linux, Docker, CI/CD, Slurm (GPU clusters), W&B, MLFlow, robot data pipelines
LLM Tooling: vLLM, LangChain, LlamaIndex, DSPy, FAISS, Langfuse

LANGUAGES

English: fluent (C1 level) | **German**: fluent (C1 level) | **Arabic**: native | **Japanese, Spanish**: beginner (A1 level)

CERTIFICATIONS

May 2024 **Industrial Robot Programming**, KUKA College, Braunschweig

Jan 2023 **Deep Learning Specialization**, Coursera

AWARDS AND HONORS

Apr 2026 – Sep 2026 **TMU Exchange Program Grant**, Tokyo Metropolitan University (TMU), Japan

Apr 2026 – Sep 2026 **Promos Scholarship**, Leibniz University Hannover, Germany

Apr 2025 – Sep 2025 **Erasmus+ Scholarship**, European Commission

Oct 2024 – Sep 2025 **Deutschlandstipendium**, Leibniz University Hannover, Germany

Apr 2018 – Sep 2023 **Government Scholarship for Academic Excellence**, Ministry of Higher Education and Scientific Research, Yemen

INTERESTS

Football and swimming | Chess | Reading | Psychology and Philosophy